

Machine Learning – COMPSCI 589 - Spring 2026

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Homework 1

1 Evaluating the k-NN Algorithm

Question 1.1: Accuracy of the k-NN model (for various values of k) when evaluated on the training set

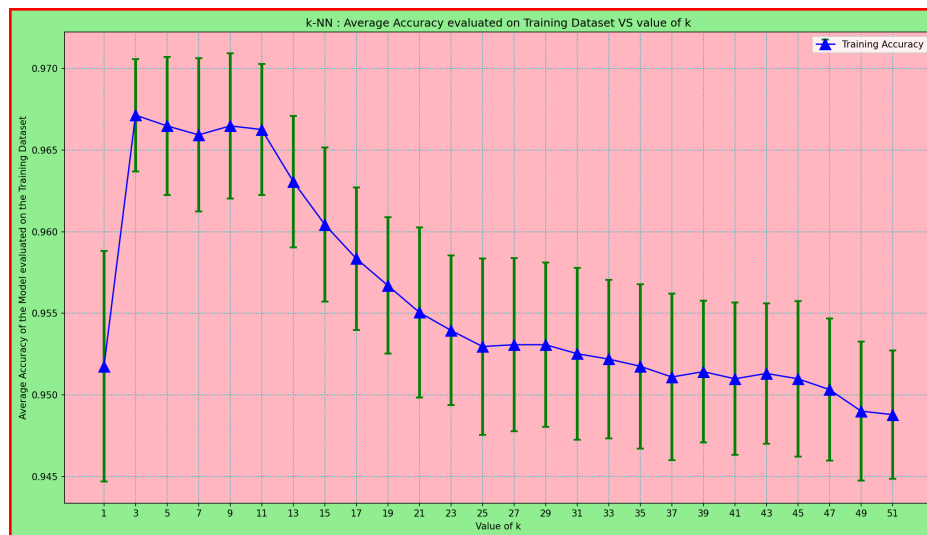


Figure 1: Accuracy of the K-NN Model - Training set

- For $K = 3$ to 11 , the K-NN algorithm model achieves optimal peak performance. The model identifies the patterns correctly and makes higher accuracy predictions. This k range captures the underlying pattern in the WDBC dataset.
- For $K = 3$ to 11 , the error bars are relatively consistent which proves the models higher accuracy predictions are more robust for varying data shuffles
- But as K increases the number of instances for comparison from the training set increases and it drives the accuracy down towards the majority class of the training dataset. Thus for $K = 51$, the accuracy of the K-NN model falls to the lowest point, because its biased more towards the majority class rather than predicting based on the pattern. For example if the WDBC dataset has a majority of instances of breast mass as benign, then with $k = 51$, the model will be more biased to predict the breast mass as benign.
- For $K = 1$, the model has lower accuracy than $K = 3$, due to models high sensitivity to outliers during accuracy calculation. If one of the neighbour is incorrect, it results in incorrect prediction
- For $K = 1$, the error bar has higher variance which proves its more sensitive to outliers.

Question 1.2: Accuracy of the k-NN model (for various values of k) when evaluated on the testing set

- For $K = 3$ to 9 , the K-NN algorithm model achieves optimal peak performance. The model identifies the patterns correctly and makes higher accuracy predictions. This k range captures the underlying pattern in the WDBC dataset. Specifically at $K = 9$ the model achieves highest accuracy
- For $K = 3$ to 9 , the error bars are relatively consistent which proves the models higher accuracy predictions are more robust for varying data shuffles

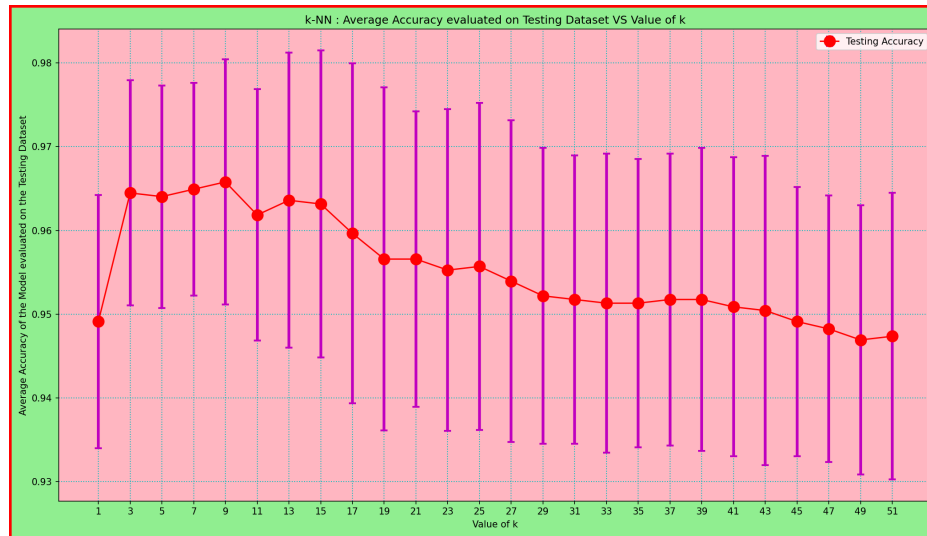


Figure 2: Accuracy of the K-NN Model - Testing set

- But as K increases the number of instances for comparison from the training set increases and it drives the accuracy down towards the majority class of the training dataset. Thus for $K = 51$, the accuracy of the K-NN model falls to the lowest point, because its biased more towards the majority class rather than predicting based on the pattern.
- For $K = 11$ to 51 , the error bars variance drastically increases which proves the model is weak and non robust for higher K values
- For $K = 1$, the model has lower accuracy than $K = 3$, due to models high sensitivity to outliers during accuracy calculation. The model is overfitting to training data. If one of the neighbour is incorrect, it results in incorrect prediction

Question 1.3:

Performance of the Training set as a function of K

- For $K = 3$ to 11 , the K-NN algorithm model achieves optimal peak performance. The model identifies the patterns correctly and makes higher accuracy predictions. This k range captures the underlying pattern in the WDBC dataset.

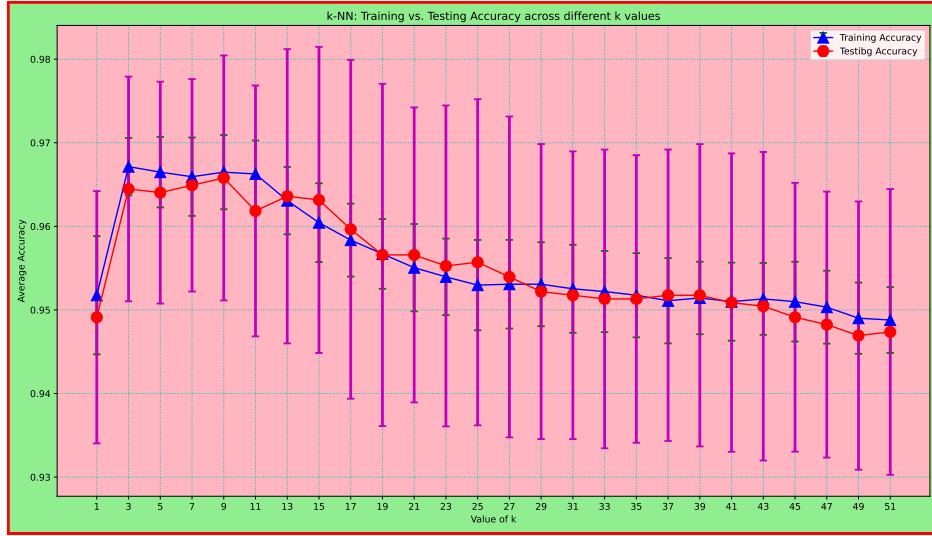


Figure 3: Accuracy of the K-NN Model - Training vs Testing set

- For $K = 3$ to 11, the error bars are relatively consistent which proves the models higher accuracy predictions are more robust for varying data shuffles
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Performance of the Testing set as a function of K

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Question 1.4:

- Overfitting - For $K = 1$, the K-NN algorithm is overfitting to the training data. At $K = 1$, the model is capturing noises and outliers, which leads to poor generalization. When compared with $K = 3$ to 9 range, the testing accuracy for $K = 1$ is lower due to models high sensitivity of the outliers in the training dataset.
- Underfitting - For $K = 13$ to 51 , the K-NN algorithm is underfitting because it performs poorly on both the training and testing data. For $K = 13$ to 51 , the accuracy starts deteriorating and becomes biased towards the majority class of the dataset rather than the specific features of the input instances

Question 1.5:

- For $K = 9$, the K-NN algorithm achieves the highest accuracy on the testing dataset while maintaining low variance. At $K = 9$, the model captures the underlying patterns correctly and makes the most robust and highest accuracy prediction for new sets of testing data.

Question 1.6:

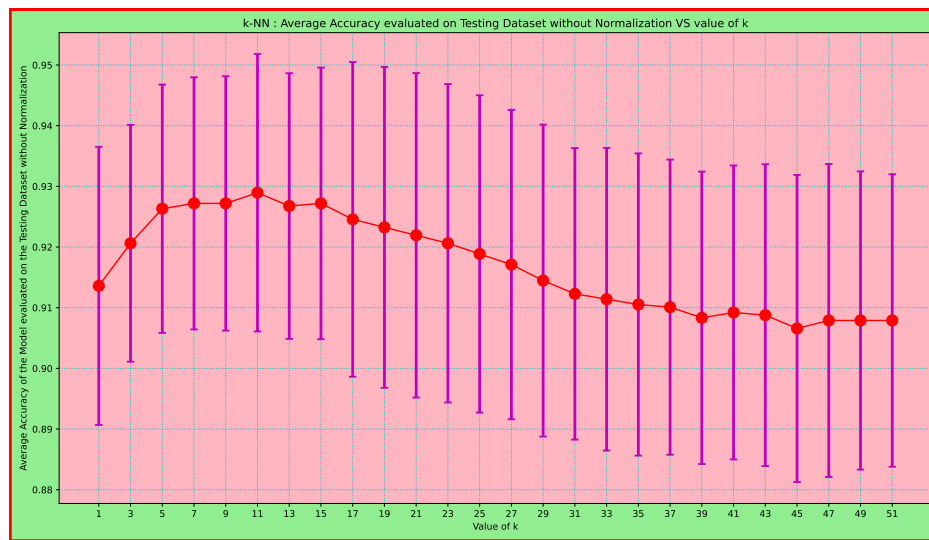


Figure 4: Accuracy of the K-NN Model - Testing set without Normalization

- Based on the result without feature normalization, the best value of K that results in K-NN performing the best on the testing dataset without normalization is $K = 11$
- The highest accuracy of the K-NN algorithm on the testing dataset with feature normalization is 0.965 for $K = 9$. In comparison, the highest accuracy of the K-NN algorithm without feature normalization is 0.929 for $K = 11$. The accuracy of the K-NN algorithm decreased due to the omission of feature normalization. The accuracy would have fallen because without normalization, attributes with larger numerical ranges contribute more significantly to the distance calculation than attributes

with smaller ranges. Consequently, these larger-scale attributes affect the model's predictions much more than smaller-scale attributes, even if the latter contain more relevant breast mass diagnostic information.

2 Evaluating the Decision Tree Algorithm

Question 2.1: Histogram : Accuracy distribution when the Decision Tree algorithm is evaluated over Training data

The histogram shows the accuracy distribution when the decision tree algorithm is evaluated over training data. The decision tree algorithm is trained 100 times by shuffling and splitting the car data set and performing an accuracy calculation.

Accuracy Calculation: Training Data

Mean Accuracy of Training Data: 1.000000

Standard Deviation of Training Data: 0.000000

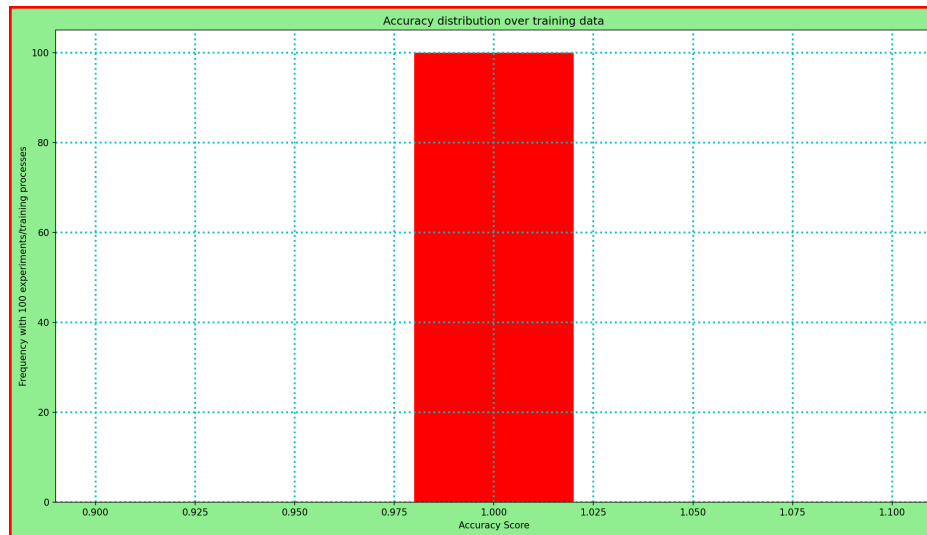


Figure 5: Frequency of observed training accuracies

Question 2.2: Histogram : Accuracy distribution when the Decision Tree algorithm is evaluated over Testing data

The histogram shows the accuracy distribution when the decision tree algorithm is evaluated over testing data. The decision tree algorithm is tested 100 times by shuffling and splitting the car data set and performing an accuracy calculation.

Accuracy Calculation: Testing Data

Mean Accuracy of Testing Data: 0.925347

Standard Deviation of Testing Data: 0.014323

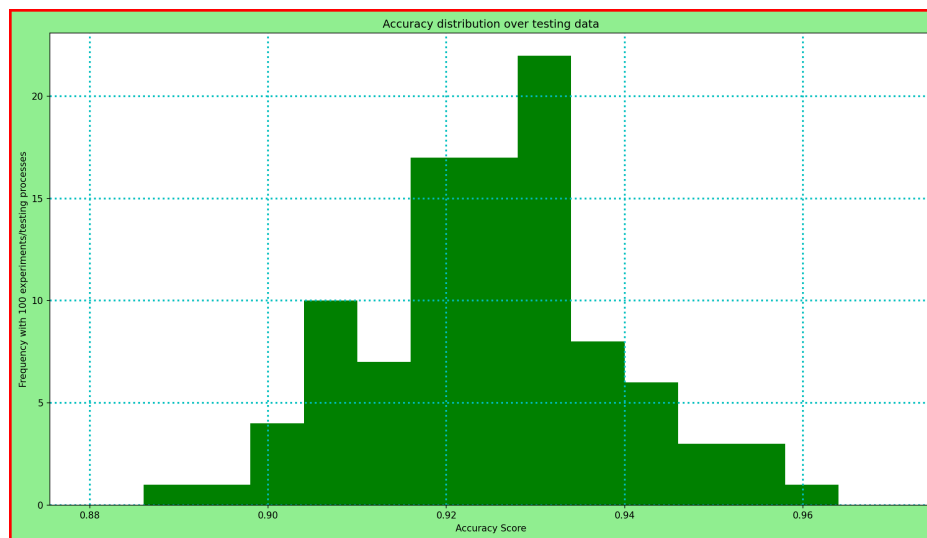


Figure 6: Frequency of observed testing accuracies

Question 2.3:

The training accuracy histogram has a single vertical bar with an accuracy score value of 1.0. Here, all the records of the car dataset are perfectly partitioned, and all partitions are as homogeneous as possible by belonging to the same class. Since all instances of the given partition/node belong to

the same class, that node becomes a leaf node labeled with that class. The tree has over-fitted the dataset. The decision tree is able to remember each feature and classify each record. As a result, the mean value is 1.0. Each record is split using the same feature and partitioned to the same class in all 100 instances of the decision tree run, resulting in a standard deviation of 0.0 for the training accuracy.

The testing accuracy histogram is more like a bell-shaped normal distribution histogram. Since we are shuffling the car dataset, the decision tree handles the dataset as a new dataset and applies the algorithms as a new application because it has not seen the data before. Due to this, there is variation in the accuracy in each run of the 100 instances of the test run. The testing histogram has significant variance. Out of the 100 shuffles, some created a testing car dataset similar to the training dataset which the algorithm is able to predict easier. But in some cases, the testing car dataset has new records which the decision tree algorithm did not face during training which caused lower accuracy. This results in significant variance and lower mean accuracy when compared to the training dataset.

Question 2.4:

In the training dataset, all instances of the given partition/node belong to the same class resulting in perfect partitioning. As a result, the training accuracy is 1.0 with no variance. When the same decision tree algorithm is applied to the testing dataset, the memorization of the features in the dataset and classification during the training process doesn't apply well to the testing dataset. This causes bell-shaped histograms with significant variance resulting in over-fitting of the decision tree algorithm. The model is highly sensitive to specific training set. The difference between zero variance in training dataset result and the bell shaped, high-variance testing dataset result of the decision tree algorithm confirms that the model memorized the training results and applied it to testing data, capturing noise and do not generalize well (make correct predictions) to new instances of the various test data.

Question 2.5:

The decision tree model learned by the machine learning algorithm is highly sensitive to small variations of the dataset. This leads to overfitting because

the model focuses too intensely on the specific details of the training dataset. Even minor change of few instances encountered during testing the testing dataset causes the machine learning algorithm to loose its ability of capturing the overall patterns in the dataset, leading to significant variances. A robust decision tree machine learning algorithm needs to be reliable and produce same and consistent result regardless of training or testing dataset. The average training accuracies of 1.0 and standard deviation of 0.0 proves that the decision tree algorithm is able to perform perfect partitioning by memorizing of the training dataset. But the average testing accuracies of 0.925347 and standard deviation of 0.014323 of the testing dataset proves that the decision tree algorithm looses its ability to capture the overall patterns due to new instances in the testing dataset. This sensitivity to new instances in the testing dataset provides empirical evidence that the Decision Tree algorithm is non-robust.